

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA0407

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

Error Analysis Task

1.A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer

Given

- Training MSE = 0.85
- Testing MSE = 5.20
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Analysis

The training error is very low, while the testing error is much higher. This indicates that the neural network performs well on the training data but poorly on unseen test data. This situation is known as overfitting.

Possible Reasons

1. Overfitting

- The model has memorized the training data instead of learning general patterns.

2. Insufficient Training Data

- A small dataset may not represent all possible student score patterns.

3. High Model Complexity

- Too many hidden layers or neurons can make the model overly complex.

4. Noisy or Poor-Quality Data

- Incorrect or inconsistent student records reduce prediction accuracy.

5. Lack of Regularization

- Without techniques like dropout or L2 regularization, the model may overfit.

Corrective Measures

- Collect more high-quality training data.
- Apply Dropout to reduce overfitting.
- Use L1/L2 Regularization.
- Perform Early Stopping during training.
- Simplify the neural network by reducing hidden layers or neurons.
- Use Cross-Validation to improve model generalization.
- Normalize and clean the dataset before training.

Interpretation

Since the Training MSE (0.85) is much lower than the Testing MSE (5.20), the model has poor generalization ability. The main issue is overfitting, and techniques such as regularization, dropout, early stopping, and collecting more training data can significantly improve the model's performance on unseen data.

Conclusion

The large difference between training and testing MSE indicates that the neural network is overfitting. By reducing model complexity, improving data quality, and using regularization techniques, the testing error can be reduced and the model's prediction accuracy can be improved.

Code of Conduct:

I CH. Durga Prasad (192425305) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided